

The interpolated stochastic heat and wave equation: solvability, exact moment asymptotics and special case properties.

by

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Abstract

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Acknowledgments

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Chapter 1

Introduction

The study of deterministic Calculus can be thought of as the starting point of the search to model physical phenomena with math. The story has it, that Issac Newton was first inspired to develop what would be known as the theory of Calculus so that he could describe a falling apple from a tree. Very simple scenarios, such as the apple falling from the tree, were for the first time successfully described but of course, not all scenarios are "simple" and more theory needed to be developed.

Consider the situation where we have a perfectly round piece of wire with infinitesimally small radius and suppose that we initially heat this wire. Assuming that there is no additional heat being added to our system, how could we model the evolution of the temperature at a specific point x on the piece of wire? In other words, can we find a function $u(t, x)$ that gives us the exact temperature at position x on the wire at time t ? The answer is yes, and thanks to Joesph Fourier, we know that this scenario can be modeled by the so called heat equation:

$$\begin{cases} \mathcal{H}u(t, x) = 0 & t \geq 0, x \in \mathbb{R} \\ u(0, x) = f(x) & x \in \mathbb{R}, \end{cases}$$

where $\mathcal{H}$ is the heat operator:

$$\mathcal{H}u(t, x) = \frac{\partial u}{\partial t}(t, x) - \frac{\partial^2 u}{\partial x^2}(t, x)$$

In order to fully utilize the above model, we need to make the problem more realistic. In any real-life scenario, we can not safely assume that there will not be any outside heat affecting our system. Moreover, it is safe to assume that any outside heat affecting our system will be

done in some sort of "random" way. The question now is, how do we model this randomness? When the spatial dimension is 1, as is above in our heat model, John Walsh laid out the framework to describe such problem in his seminal textbook, *An Introduction to Stochastic Partial Differential Equations*. The idea is to construct a *Gaussian space-time white noise*, $\dot{W}$ (see Section 1.1), and consider the following *stochastic heat equation*:

$$\begin{cases} \mathcal{H}u(t, x) = \rho(u(t, x))\dot{W}(t, x) & t \geq 0, x \in \mathbb{R} \\ u(0, x) = f(x) & x \in \mathbb{R}, \end{cases} \quad (1.0.1)$$

where ρ is a globally Lipschitz function and f describes the initial heat of our wire. See Section 1.2 for a through introduction of this set up.

This notation is purely formal as $\dot{W}$ does not exist as a function but as a distribution. However we can make sense of the above equation as the following *stochastic Walsh integral equation*:

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y)f(y)dy + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y)\rho(u(s, y))W(ds, dy), \quad (1.0.2)$$

where $G(t, x)$ is the Gaussian heat kernel:

$$G(t, x) = \frac{\exp\left(-\frac{|x|^2}{2t}\right)}{\sqrt{2\pi t}}, \quad t > 0, x \in \mathbb{R}.$$

Thus through this novel idea due to Walsh, we now move one step closer to being able to model real-life phenomena under much more realistic conditions. The next question then would be under what conditions is the equation (1.0.1) well-posed? Also, can we apply this theory to other partial differential operators such as the wave operator,

$$\mathcal{W}u(t, x) = \frac{\partial^2 u}{\partial t^2}(t, x) - \frac{\partial^2 u}{\partial x^2}(t, x)$$

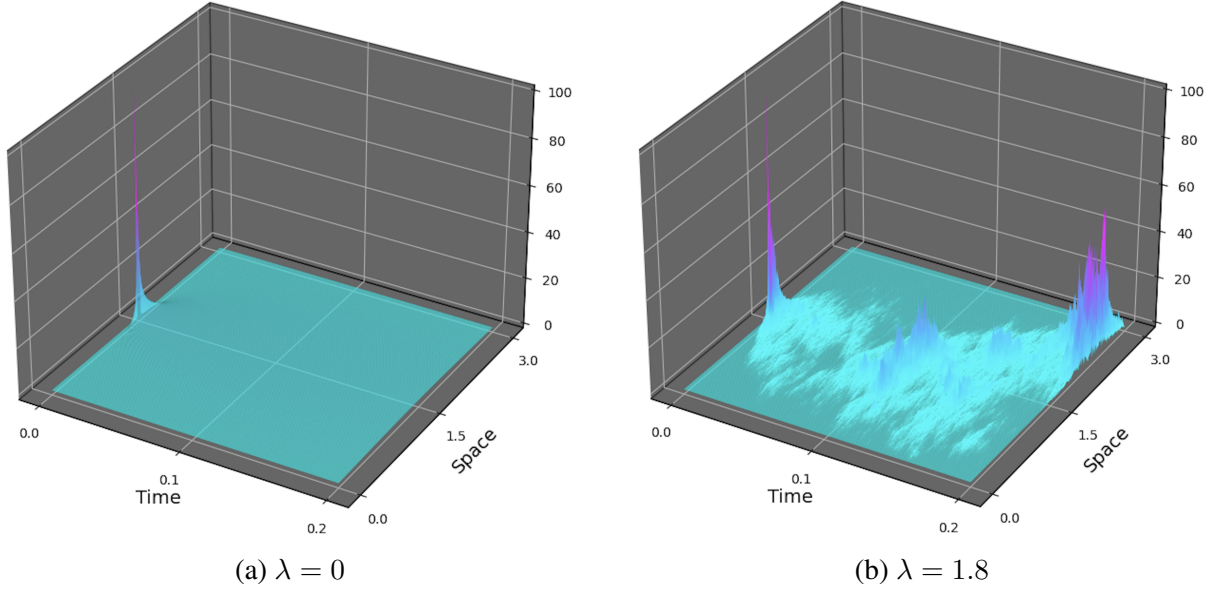


Figure 1.1: Simulations of Equation (1.0.1) with $f(\cdot) = \delta_0(\cdot)$ and $\rho(u) = \lambda u$.

or even a more general fractional operator of the following type:

$$\mathcal{L}_{b,\alpha}u(t, x) = \partial_b u(t, x) - (-\Delta)^{-\alpha/2}u(t, x),$$

where ∂_b is the *Caputo* time-derivative of order $b > 0$ and $(-\Delta)^{-\alpha/2}$ is the fractional Laplacian. The fractional operator, $\mathcal{L}_{\alpha,\beta}$ will be thoroughly discussed in section 1.3 below as well as in Chapters 3 and 4. As for the stochastic wave equation, we direct the interested reader to [Dal+09; MS21; RC21] to name a few.

We will finish this introduction with a brief discussion about the stochastic heat equation before giving a more thorough treatment in Section 1.2. As for the well-posedness of Equation (1.0.1) above, Chen and Dalang [CD15] proved the following:

Theorem 1.0.1. *Suppose that the function ρ is globally Lipschitz function and the initial distribution $f(\cdot)$ satisfies the following:*

$$|\rho(x)| \leq L_\rho(L_0^2 + |x|^2), \quad \text{and} \quad \int_{\mathbb{R}} \exp(-a|x|^2) f(dx) < \infty,$$

for some $L_b, L_0 > 0$ and any $a > 0$. Then for $p \geq 2$, there exists a unique, $L^p(\Omega)$ -continuous process that almost surely satisfies (1.0.2).

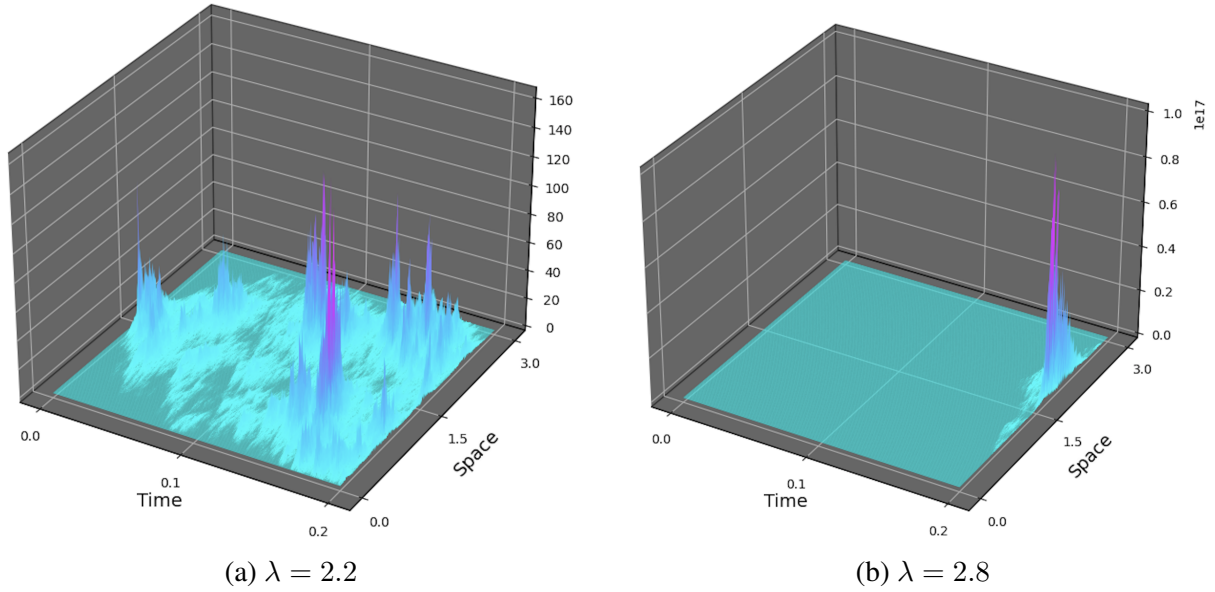


Figure 1.2: Simulations of Equation (1.0.1) with $f(\cdot) = \delta_0(\cdot)$ and $\rho(u) = \lambda u$.

It is important to note that the above theorem allows for the initial condition to be the Dirac delta distribution. Moreover, this is a very natural choice and has many real world applications.

The simulations in Figures 1.1 and 1.2 illustrate a phenomena not seen in the deterministic case, that is, the formation of very large peaks widely separated by very deep valleys. This may seem unrealistic in practice but there is in fact a situation where one may make sense of this. Consider the $\delta_0(\cdot)$ initial condition to be representative of the "big bang" at the creation of our universe. One may assume that this is a highly random and chaotic event and that the transfer of heat may be roughly modeled Equation (1.0.1) with $f(\cdot) = \delta_0(\cdot)$ and $\rho(u) = \lambda u$ for some "large" $\lambda > 0$. These tall, but isolated, peaks that form may be thought of as the formation of galaxies and the empty space in between can just be thought of as the emptiness of outer space. Of course this is highly simplified, but still a nice way to picture what is happening in the simulations.

Describing the growth of the peaks is an interesting question on its own. At the moment, it is an open question to describe the growth of the sample paths in time to the stochastic heat equation, in other words, the growth of the function

$$t \mapsto u(t, x)(\omega), \quad x \in \mathbb{R}^d, \omega \in \Omega.$$

However, we do have some results on the growth in time of the p^{th} moments of the solution for $p \geq 2$. Some references that address this matter can found in [CD15; CH19; CK19] and also Chapter 3 below. Namely, in many cases of interest, the p^{th} moments to the solution of the stochastic heat equation, and more generally the space-time fractional equation defined in Section 1.3, grow exponentially in time. This phenomena is referred to as *intermittency*.

1.1 Gaussian Noise

The driving force of a stochastic partial differential equation is the noise term, usually denoted by $\dot{W}$. Throughout this thesis we will consider three such types: white in time and white in space, white in time and colored in space, and time independent. We will now define each of the three.

1.1.1 White in time and either spatially white or colored

We follow the set up as in [Dal+09]. First, let $\Gamma(dx)$ be a non-negative and non-negative definite tempered measure, ie,

$$\int_{\mathbb{R}^d} \Gamma(dx) (\phi * \tilde{\phi})(x) \geq 0, \quad \forall \phi \in \mathcal{L}(\mathbb{R}^d),$$

where $\mathcal{L}(\mathbb{R}^d)$ denotes the space of Schwartz functions and $\tilde{\phi}(x) = \phi(-x)$ and in addition

$$\int_{\mathbb{R}^d} \Gamma(dx) \frac{1}{(1 + |x|^2)^r} < \infty.$$

The Bochner-Schwartz theorem [Sch66] guarantees the existence of a measure μ , referred to as the *spectral measure*, whose Fourier transform is Γ : $\mathcal{F}\mu = \Gamma$. This means that

$$\int_{\mathbb{R}^d} \Gamma(dx) \phi(x) = \int_{\mathbb{R}^d} \mu(d\xi) \mathcal{F}\phi(\xi),$$

where we choose the following as our definition of the Fourier transform:

$$\mathcal{F}\phi(\xi) = \int_{\mathbb{R}^d} dx \exp(-i\xi \cdot x) \phi(x).$$

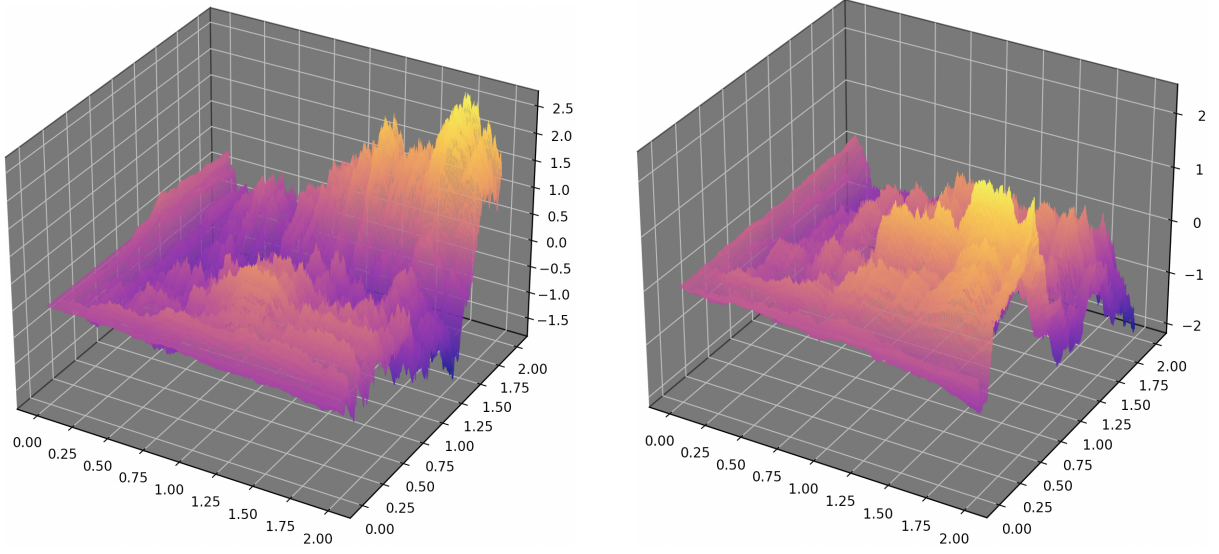


Figure 1.3: A simulation of two sample paths of a Brownian sheet.

Definition 1.1.1. Consider the $L^2(\Omega, \mathcal{F}, \mathbb{P})$ valued mean zero Gaussian process

$$W = (W(\phi), \phi \in C_0^\infty(\mathbb{R}^{d+1}))$$

such that

$$\mathbb{E}(W(\phi)W(\psi)) = \int_0^\infty ds \int_{\mathbb{R}^{2d}} \Gamma(dx) \left(\phi(s, \cdot) * \tilde{\psi}(s, \cdot) \right) (x).$$

When $\Gamma(dx) = \delta_0(x)$ then W is referred to as a *spatially homogeneous Gaussian space-time white noise*. Otherwise we say that W is white in time and *colored* in space.

Proposition 1.1.2 (Proposition 4.19 [DZ92]). A *space-time white noise* is the distributional derivative of a Brownian sheet. See Figure 1.3.

Note that in the case of a space-time white noise that the covariance takes the following form:

$$\mathbb{E}(W(\phi)W(\psi)) = \int_0^\infty ds \int_{\mathbb{R}^d} dx \phi(s, x)\psi(s, x),$$

and when the noise is colored and Γ has density f then the covariance takes the form

$$\mathbb{E}(W(\phi)W(\psi)) = \int_0^\infty ds \int_{\mathbb{R}^{2d}} dx dy \phi(s, x)f(x - y)\psi(s, y). \quad (1.1.1)$$

Lastly, by the convolution property of the Fourier transform, we may say that

$$\mathbb{E}(W(\phi)W(\psi)) = \int_0^\infty ds \int_{\mathbb{R}^d} \mu(d\xi) \mathcal{F}\phi(s, \xi) \overline{\mathcal{F}\psi(s, \xi)}.$$

Definition 1.1.3. Following [Wal86], we extend W to a sigma-finite $L^2(\Omega)$ -valued measure $A \mapsto W(A)$ where A is a bounded Borel subset of $(0, \infty) \times \mathbb{R}^d$. We then define the *natural filtration*, $(\mathcal{F}_t)_{t \geq 0}$, where

$$\mathcal{F}_t = \sigma(W_s(A) : 0 \leq s \leq t, A \in \mathcal{B}_b((0, \infty) \times \mathbb{R}^d)) \vee \mathcal{N},$$

where $\mathcal{N}$ denotes the set of $\mathbb{P}$ -null sets in $\mathcal{F}$ and $\mathcal{B}_b((0, \infty) \times \mathbb{R}^d)$ is the space of bounded Borel subsets of $(0, \infty) \times \mathbb{R}^d$.

1.1.2 Time independent

A time independent Gaussian noise is defined similarly as above.

Definition 1.1.4. A *time-independent spatially homogeneous* Gaussian noise is a $L^2(\Omega, \mathcal{F}, \mathbb{P})$ valued mean zero Gaussian processes indexed on $C_0^\infty(\mathbb{R}^d)$,

$$W = (W(\phi) : \phi \in C_0^\infty(\mathbb{R}^d)),$$

with the following covariance structure

$$\mathbb{E}(W(\phi)W(\psi)) = \int_{\mathbb{R}^d} \mu(d\xi) \mathcal{F}\phi(\xi) \overline{\mathcal{F}\psi(\xi)},$$

where we have the relation that $\Gamma = \mathcal{F}\mu$ just as we did in Section 1.1.1.

1.2 The Stochastic Heat Equation

We will now introduce more formally the stochastic heat equation along with several important results that will be needed throughout this thesis. There many resources to choose for an introduction to the stochastic heat equation [CD15; CK19; Dal+09; DZ92; Kho14; Wal86].

Recall from (1.0.1) that the stochastic heat equation can be written informally in the following way:

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) = \frac{\nu}{2} \Delta_x u(t, x) + \rho(u(t, x)) \dot{W}(t, x) & t \geq 0, x \in \mathbb{R}^d \\ u(0, x) = \mu(\cdot) & x \in \mathbb{R}^d. \end{cases} \quad (1.2.1)$$

The initial distribution μ is a deterministic and locally finite Borel measure which must satisfy the following inequality

$$\int_{\mathbb{R}^d} \exp(-a|x|^2) |\mu|(dx) < \infty, \quad \text{for all } a > 0, \quad (1.2.2)$$

where $|\mu| = \mu^+ + \mu^-$. In general, $d \geq 1$, $\nu > 0$ and the noise may be either of the three noises introduced in Section 1.1. However, as we will see in Chapter 2, our only case of interest is when $d \geq 3$ and the noise is white in time and colored in space. For that reason we will restrict ourselves to that case.

We will denote by $\hat{f}(d\xi)$, the spectral measure of the noise $\dot{W}$ and in the case when Γ has density f , as in (1.1.1) above, then the spectral measure takes the form $\hat{f}(\xi)d\xi$. We define the following property known as *Dalang's condition*:

$$\Upsilon(\beta) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty, \quad \text{for some and hence all } \beta > 0. \quad (1.2.3)$$

By a solution to (1.2.1), we mean a solution to the following stochastic integral equation:

$$\begin{aligned} u(t, x) &= \int_0^\infty G(t, x - y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y) \rho(u(s, y)) W(ds, dy) \\ &= J_0(t, x) + I(t, x) \end{aligned} \quad (1.2.4)$$

where the above is a *Walsh* stochastic integral [Wal86] and G is the heat kernel

$$G(t, x) = (2\pi\nu t)^{-d/2} \exp\left(-\frac{|x|^2}{2\nu t}\right).$$

Definition 1.2.1. A random field $u = (u(t, x) : t \geq 0, x \in \mathbb{R}^d)$ is a *mild solution* to (1.2.1) if

1. u is adapted to the filtration generated by the noise W ,
2. u is jointly measurable with respect to $\mathcal{B}((0, \infty) \times \mathbb{R}^d) \times \mathcal{F}$,
3. $\|I(t, x)\|_2 < \infty$ for all $(t, x) \in (0, \infty) \times \mathbb{R}^d$,
4. u satisfies (1.2.4) a.s. for all $(t, x) \in (0, \infty) \times \mathbb{R}^d$.

We have the following solvability theorem

Theorem 1.2.2 (Theorem 2.4 [CK19]). *Suppose that the initial distribution and the spectral measure satisfy both (1.2.2) and (1.2.2) respectively. Suppose that ρ is globally Lipschitz with Lipschitz constant L_ρ . Then there exists a unique (in the sense of versions) random-field solution starting from μ that is $L^2(\Omega)$ -continuous.*

The next result gives us a very useful bound on the p^{th} moments of the solution. In order to fully utilize the result, we need to strengthen some of our assumption to the following:

$$\Upsilon(0) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{|\xi|^2} < \infty \quad \text{and} \quad 64L_b^2 < \frac{1}{4\Upsilon(0)}; \quad (1.2.5)$$

Theorem 1.2.3 (Theorem 1.7 [CH19], Lemma 2.5 [CK19]). *Under the assumptions of Theorem 1.2.2, for any $t > 0$ and $x \in \mathbb{R}^d$, the solution to (1.2.1), $u(t, x)$, is in $L^p(\Omega)$ and for $p \geq 2$*

$$\|u(t, x)\|_p \leq [\bar{\varsigma} + \sqrt{2}(G(t, \cdot) * |\mu|)(x)] H(t; \gamma_p)^{1/2}, \quad (1.2.6)$$

where $\bar{\varsigma} = |\rho(0)|/L_\rho$, $\gamma_p = 32pL_\rho^2$ and the function $H(t; \gamma_p)$ is nondecreasing in t (see [CH19] for the expression of H). Moreover, if two conditions in (1.2.5) hold, then this function $H(t; \gamma_p)$ is bounded in t . Hence, by choosing $C = (\sqrt{2} \vee \bar{\varsigma}) \sup_{t \geq 0} H(t; \gamma_p)^{1/2} < \infty$, we have that

$$\|u(t, x)\|_p \leq C \left(1 + (G(t, \cdot) * |\mu|)(x) \right). \quad (1.2.7)$$

1.3 The Interpolated Stochastic Heat and Wave Equation

Chapter 2

Invariant Measures for the Stochastic Heat Equation

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Chapter 3

Exact Moment Asymptotics for the Interpolated Stochastic Heat and Wave Equation

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Chapter 4

Existence of Global Solutions to the ISHWE with a Super-linear Diffusion Term

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Appendices

Appendix A

Notes on the style-file project

These style-files have been modified by Christopher Wilson to support the new Electronic Thesis and Dissertation process. The following was written before the ETD guidelines update on Fall 2009. Changes to the original appendix have been shown in *italics*.

These style-files for use with $\text{\LaTeX}$ *were* maintained by Darrel Hankerson¹ and Ed Slaminka².

In 1990, department heads and other representatives met with Dean Doorenbos and Judy Bush-Crofton (then responsible for manuscript approval). This meeting was prompted by a memorandum³ from members of the mathematics departments concerning the *Thesis and Dissertation Guide* and the approval process. There was wide agreement among the participants (including Dean Doorenbos) to support the basic recommendations outlined in the memorandum. The revised *Guide* reflected some (but not all) of the agreements of the meeting.

Ms Bush-Crofton was supportive of the plan to obtain “official approval” of these style files.⁴ Unfortunately, Ms Bush-Crofton left the Graduate School before the process was completed. In 1994, we were revisiting some of the same problems which were resolved at the 1990 meeting.

In Summer 1994, I sent several memoranda to Ms Ilga Trend of the Graduate School, reminding her of the agreements made at the 1990 meeting. Professors A. Scottedward Hodel and Stan Reeves provided additional support. In short, it is essential that the Graduate School honor its commitments of the 1990 meeting. It should be emphasized that Dean Doorenbos is to thank for the success of that meeting.

Maintaining these $\text{\LaTeX}$ files has been more work than expected, in part due to continuing changes in requirement by the graduate school. The Graduate School occasionally has complete memory loss about the agreements of the 1990 meeting. If the Graduate School rejects your manuscript based on items controlled by the style-files, ask your advisor to contact the Graduate school (and copy to me) to urge cooperation.

Finally, there have been several requests for additions to the package (mostly formatting changes for figures, etc.). While such changes are not really part of the thesis-style package, it could be beneficial to collect these options and distribute with the package (making it easier on the next student). I’m especially interested in changes needed by various departments.

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³Originally, the memorandum was presented to Professor Larry Wit. A copy is available on request.

⁴Followup memoranda gave a definition of “official approval.” Copies will be sent on request.